



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen



CSC 6011 - Theory of Computation - Lecture 7: Decision Problems for Automata and Grammars

Time: Mon & Wed, 4:00 - 5:20 PM

Location: Teaching A, 107

Instructor: Tao LIN (林涛)

TA: Yefan GAO (高叶繁)

Fall 2026

Lecture 7

Last time:

- Equivalence of variants of the Turing machine model
 - a. Multi-tape TMs
 - b. Nondeterministic TMs
 - c. Enumerators
- Church-Turing Thesis
- Notation for encodings and TMs

Today:

- Decision problems for automata and grammars

TMs and Encodings – review

A TM has 3 possible outcomes for each input w :

1. Accept w (enter q_{acc})
2. Reject w by halting (enter q_{rej})
3. Reject w by looping (running forever)

A is T-recognizable if $A = L(M)$ for some TM M .

A is T-decidable if $A = L(M)$ for some TM decider M (halts on all inputs)

$\langle O_1, O_2, \dots, O_k \rangle$ encodes objects $O_1, O_2, \dots, O_k$ as a single string.

Notation for writing a TM M is

$M =$ “On input w
[English description of the algorithm]”

Acceptance Problem for DFAs

Let $A_{\text{DFA}} = \{ \langle B, w \rangle \mid B \text{ is a DFA and } B \text{ accepts } w \}$

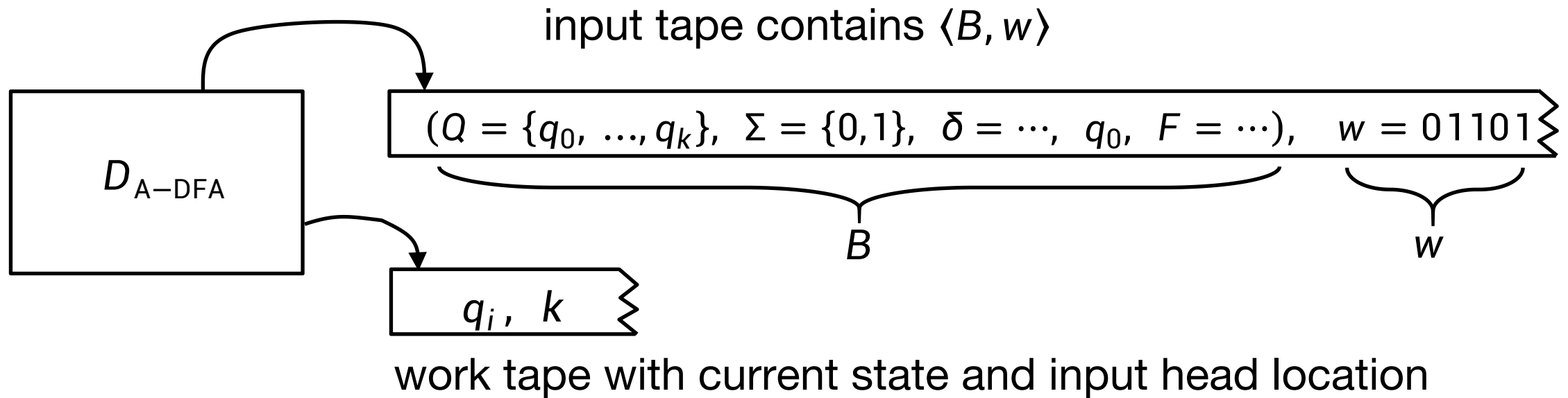
Theorem: A_{DFA} is decidable

Proof: Give TM $D_{\text{A-DFA}}$ that decides A_{DFA} .

$D_{\text{A-DFA}}$ = “On input s

1. Check that s has the form $\langle B, w \rangle$ where B is a DFA and w is a string; *reject* if not.
2. Simulate the computation of B on w .
3. If B ends in an accept state then *accept*.
If not then *reject*.”

Acceptance Problem for DFAs



Acceptance Problem for DFAs

Let $A_{\text{DFA}} = \{ \langle B, w \rangle \mid B \text{ is a DFA and } B \text{ accepts } w \}$

Theorem: A_{DFA} is decidable

Proof: Give TM $D_{\text{A-DFA}}$ that decides A_{DFA} .

$D_{\text{A-DFA}}$ = “On input s

1. Check that s has the form $\langle B, w \rangle$ where B is a DFA and w is a string; *reject* if not.
2. Simulate the computation of B on w .
3. If B ends in an accept state then *accept*.
If not then *reject*.”

Shorthand:

On input $\langle B, w \rangle$

Acceptance Problem for NFAs

Let $A_{\text{NFA}} = \{ \langle B, w \rangle \mid B \text{ is an NFA and } B \text{ accepts } w \}$

Theorem: A_{NFA} is decidable

Proof: Give TM $D_{\text{A-NFA}}$ that decides A_{NFA} .

$D_{\text{A-NFA}} =$ “On input $\langle B, w \rangle$

1. Convert NFA B to equivalent DFA B' .
2. Run TM $D_{\text{A-DFA}}$ on input $\langle B', w \rangle$. [Recall that $D_{\text{A-DFA}}$ decides A_{DFA}]
3. *Accept* if $D_{\text{A-DFA}}$ accepts.
Reject if not.”

New element: Use conversion construction and previously constructed TM as a subroutine.

Emptiness Problem for DFAs

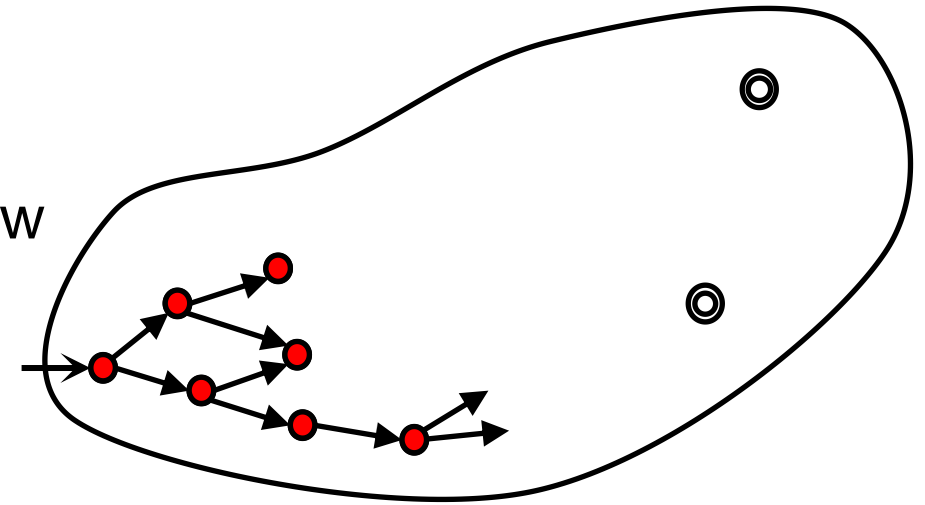
Let $E_{\text{DFA}} = \{ \langle B \rangle \mid B \text{ is a DFA and } L(B) = \emptyset \}$

Theorem: E_{DFA} is decidable

Proof: Give TM $D_{E-\text{DFA}}$ that decides E_{DFA} .

$D_{E-\text{DFA}} =$ “On input $\langle B \rangle$ [IDEA: Check for a path from start to accept.]

1. **Mark** start state.
2. Repeat until no new state is marked:
Mark every state that has an incoming arrow from a previously marked state.
3. *Accept* if no accept state is marked.
Reject if some accept state is marked.”



“Breadth-first search”

Equivalence problem for DFAs

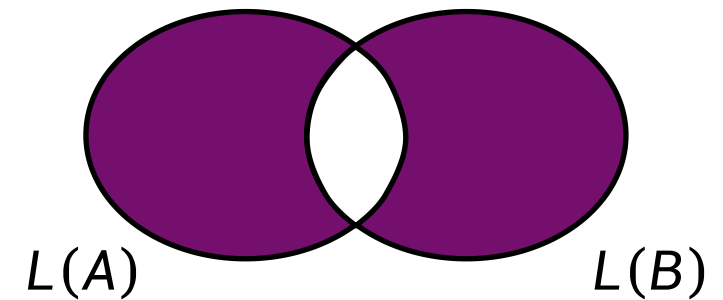
Let $EQ_{DFA} = \{ \langle A, B \rangle \mid A \text{ and } B \text{ are DFAs and } L(A) = L(B) \}$

Theorem: EQ_{DFA} is decidable

Proof: Give TM D_{EQ-DFA} that decides EQ_{DFA} .

D_{EQ-DFA} = “On input $\langle A, B \rangle$ [IDEA: Make DFA C that accepts w where A and B disagree.]

1. Construct DFA C where $L(C) = (L(A) \cap \overline{L(B)}) \cup (\overline{L(A)} \cap L(B))$.
2. Run D_{E-DFA} on $\langle C \rangle$.
3. *Accept* if D_{E-DFA} accepts.
Reject if D_{E-DFA} rejects.”



Symmetric difference

Equivalence problem for DFAs

Let $EQ_{DFA} = \{ \langle A, B \rangle \mid A \text{ and } B \text{ are DFAs and } L(A) = L(B) \}$

Theorem: EQ_{DFA} is decidable

Proof: Give TM D_{EQ-DFA} that decides EQ_{DFA} .

$D_{EQ-DFA} =$ “On input $\langle A, B \rangle$ [IDEA: Make DFA C that accepts w where A and B disagree.]

Mini-Quiz 7.1

Let $EQ_{REG} = \{ \langle R_1, R_2 \rangle \mid R_1 \text{ and } R_2 \text{ are regular expressions and } L(R_1) = L(R_2) \}$. Can we now conclude that EQ_{REG} is decidable?

- a) Yes, it follows immediately from things we’ve already shown.
- b) Yes, but it would take significant additional work.
- c) No, intersection is not a regular operation.

Acceptance Problem for CFGs

Let $A_{\text{CFG}} = \{ \langle G, w \rangle \mid G \text{ is a CFG and } w \in L(G) \}$

Theorem: A_{CFG} is decidable

Proof: Give TM $D_{A-\text{CFG}}$ that decides A_{CFG} .

Problem: Derivation of w can be very long...

Chomsky Normal Form

Defn: Chomsky Normal Form (CNF) only allows rules:

$$A \rightarrow BC$$

$$B \rightarrow b$$

$S \rightarrow \varepsilon$ is also allowed if the language contains ε .

Start variable does not occur on right-hand sides.

Lemma 1: CFG can be converted into CNF.

Proof and construction in Sipser's book.

Lemma 2: If H is in CNF and $w \in L(H)$ and $w \neq \varepsilon$, then every derivation of w has exactly $2|w| - 1$ steps.

Proof: exercise.

Example:

a CFG not in CNF:

$$S \rightarrow aS \mid b$$

It generates $L = \{b, ab, aab, aaab, \dots\}$

an equivalent CFG in CNF:

$$S_0 \rightarrow AB \mid b$$

$$B \rightarrow AB \mid b$$

$$A \rightarrow a$$

Acceptance Problem for CFGs

Let $A_{\text{CFG}} = \{ \langle G, w \rangle \mid G \text{ is a CFG and } w \in L(G) \}$

Theorem: A_{CFG} is decidable

Proof: Give TM $D_{A-\text{CFG}}$ that decides A_{CFG} .

$D_{A-\text{CFG}} =$ “On input $\langle G, w \rangle$

1. Convert G into CNF.
2. For nonempty w , try all derivations of length $2|w| - 1$.
3. *Accept* if any generate w .
Reject if not.”

If $w = \varepsilon$, accept exactly when the CNF grammar has the rule $S \rightarrow \varepsilon$.

Acceptance Problem for CFGs

Let $A_{CFG} = \{ \langle G, w \rangle \mid G \text{ is a CFG and } w \in L(G) \}$

Theorem: A_{CFG} is decidable

Proof: Give TM D_{A-CFG} that decides A_{CFG} .

Corollary: Every CFL is decidable.

Proof: Let A be a CFL, generated by CFG G .

Construct TM $M_G =$ “on input w

1. Run D_{A-CFG} on $\langle G, w \rangle$.
2. *Accept* if D_{A-CFG} accepts
Reject if it rejects.”

Mini-Quiz 7.2

Is A_{PDA} decidable?

- a) Yes.
- b) No, PDAs may be nondeterministic.
- c) No, PDAs may not halt.

Emptiness Problem for CFGs

Let $E_{\text{CFG}} = \{ \langle G \rangle \mid G \text{ is a CFG and } L(G) = \emptyset \}$

Theorem: E_{CFG} is decidable

Proof: $D_{E-\text{CFG}} =$ “On input $\langle G \rangle$ [IDEA: work backwards from terminals]

1. **Mark** all occurrences of terminals in G .
2. Repeat until no new variables are marked
Mark all occurrences of variable A if
 $A \rightarrow B_1 B_2 \cdots B_k$ is a rule and all B_i were already marked.
3. *Reject* if the start variable is marked.
Accept if not.”

$S \rightarrow R T a$

$R \rightarrow T b$

$T \rightarrow a$

Equivalence Problem for CFGs

Let $EQ_{CFG} = \{ \langle G, H \rangle \mid G, H \text{ are CFGs and } L(G) = L(H) \}$

Theorem: EQ_{CFG} is NOT decidable

Proof: Will do.

Let $AMBIG_{CFG} = \{ \langle G \rangle \mid G \text{ is an ambiguous CFG} \}$

Theorem: $AMBIG_{CFG}$ is NOT decidable

Proof: Homework.

Equivalence Problem for CFGs

Let $EQ_{CFG} = \{ \langle G, H \rangle \mid G, H \text{ are CFGs and } L(G) = L(H) \}$

Theorem: EQ_{CFG} is NOT decidable

Proof: Will do.

Mini-Quiz 7.3

Let $AMBIG_{CFG}$

Why can't we use the same technique we used to show EQ_{DFA} is decidable to show that EQ_{CFG} is decidable?

Theorem: /

Proof: Hon

- a) Because CFGs are generators and DFAs are recognizers.
- b) Because CFLs are closed under union.
- c) Because CFLs are not closed under complementation and intersection.

Acceptance Problem for TMs

Let $A_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ accepts } w \}$

Theorem: A_{TM} is not decidable

Proof: Next lecture.

Theorem: A_{TM} is T-recognizable

Proof: The following TM U recognizes A_{TM}

U = “On input $\langle M, w \rangle$

1. Simulate M on input w .
2. *Accept* if M halts and accepts.
3. *Reject* if M halts and rejects.
4. ~~*Reject* if M never halts.~~

Not a legal TM action.

Turing's original
“Universal Computing Machine”



Von Neumann said U inspired the concept of a stored program computer.

Quick review of today

1. We showed the decidability of various problems about automata and grammars: A_{DFA} , A_{NFA} , E_{DFA} , EQ_{DFA} , A_{CFG} , E_{CFG}
2. We showed that A_{TM} is T-recognizable.
3. The followings are not T-decidable: EQ_{CFG} , $\text{AMBIG}_{\text{CFG}}$, A_{TM}

These slides are adapted from Michael Sipser, *18.404J Theory of Computation*, Fall 2020. Massachusetts Institute of Technology:

- Source: MIT OpenCourseWare <https://ocw.mit.edu>

Except for separately identified third-party material, this adapted presentation is licensed under [CC BY-NC-SA 4.0](https://creativecommons.org/licenses/by-nc-sa/4.0/).